

1(a)(i)	haem ; A heme / prosthetic group I iron / iron ion / Fe / Fe²⁺ I porphyrin ring	1
1(a)(ii)	<i>one from</i> combines with / binds / carries / transports / AW, oxygen (in lungs) ; R forms bonds releases / AW, oxygen, in tissues / at low oxygen concentrations ; allows haemoglobin to transport oxygen ;	1
1(b)	<i>two from</i> 1 spherical / ball-like / rounded / AW ; R circular / round ignore 3D shape 2 (water) soluble / forms H bonds with water ; 3 hydrophilic R-groups on outside of molecule / hydrophobic R-groups on inside ; R <i>ref. to</i> 'tails' 4 dynamic / metabolic / physiological / AW, function ;	2
1(c)	made of <u>amino acids</u> ; <i>one from</i> joined by peptide bonds ; R 'peptide bond between two amino acids' R 'dipeptide bond(s)' macromolecule / long-chain (molecule) / large molecule ; repeated / many, (sub-)units / monomers ;	2

Question	Answer	Marks
1(d)	<i>three from</i> 1 (R-group of) glutamic acid / glu, is polar / hydrophilic and (R-group of) valine / val, is non-polar / hydrophobic ; 2 change in tertiary structure ; A 'change in globular shape / less globular in shape' 3 change in quaternary structure of haemoglobin ; R of β-globin 4 <i>ref. to</i> haemoglobin forms fibres (with other Hb) ; A 'sticky molecules' 5 haemoglobin is less (water) soluble ; I insoluble 6 haemoglobin is less efficient at, binding / transporting, oxygen ; A less oxyhaemoglobin A haemoglobin / β-globin, has lower affinity for oxygen A reduced oxygen carrying capacity I haemoglobin, does not / cannot, bind oxygen 7 AVP ;	3

Question	Answer	Marks
1(e)	<i>three from</i> 1 haemoglobin combines with carbon dioxide ; 2 carbon dioxide reacts with (terminal), amine group(s) / -NH_2 / -NH ; 3 to form <u>carbaminohaemoglobin</u> ; R carboxyhaemoglobin / carbonylhaemoglobin 4 each polypeptide can carry a molecule of carbon dioxide / haemoglobin can carry four molecules of carbon dioxide ; 5 carbon dioxide remains, bound / AW, to Hb until in, region of low $p\text{CO}_2$ or high $p\text{O}_2$ / pulmonary circulation / lungs / alveoli ; 6 <i>ref. to</i> (carbonic acid dissociates to form) hydrogen ions, which bind to / AW, haemoglobin ; A H^+ forms haemoglobinic acid or HHb I hydrogen unqualified 7 AVP ; e.g. hydrogencarbonate ions to plasma	3